भारतीय सूचना प्रौद्योगिकी संस्थान, पुणे
Indian Institute of Information Technology Pune
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B.Tech Project Presentation

FPGA-Accelerated Standalone Echocardiography System for Left Ventricle Segmentation and Ejection Fraction Estimation

Team Members:

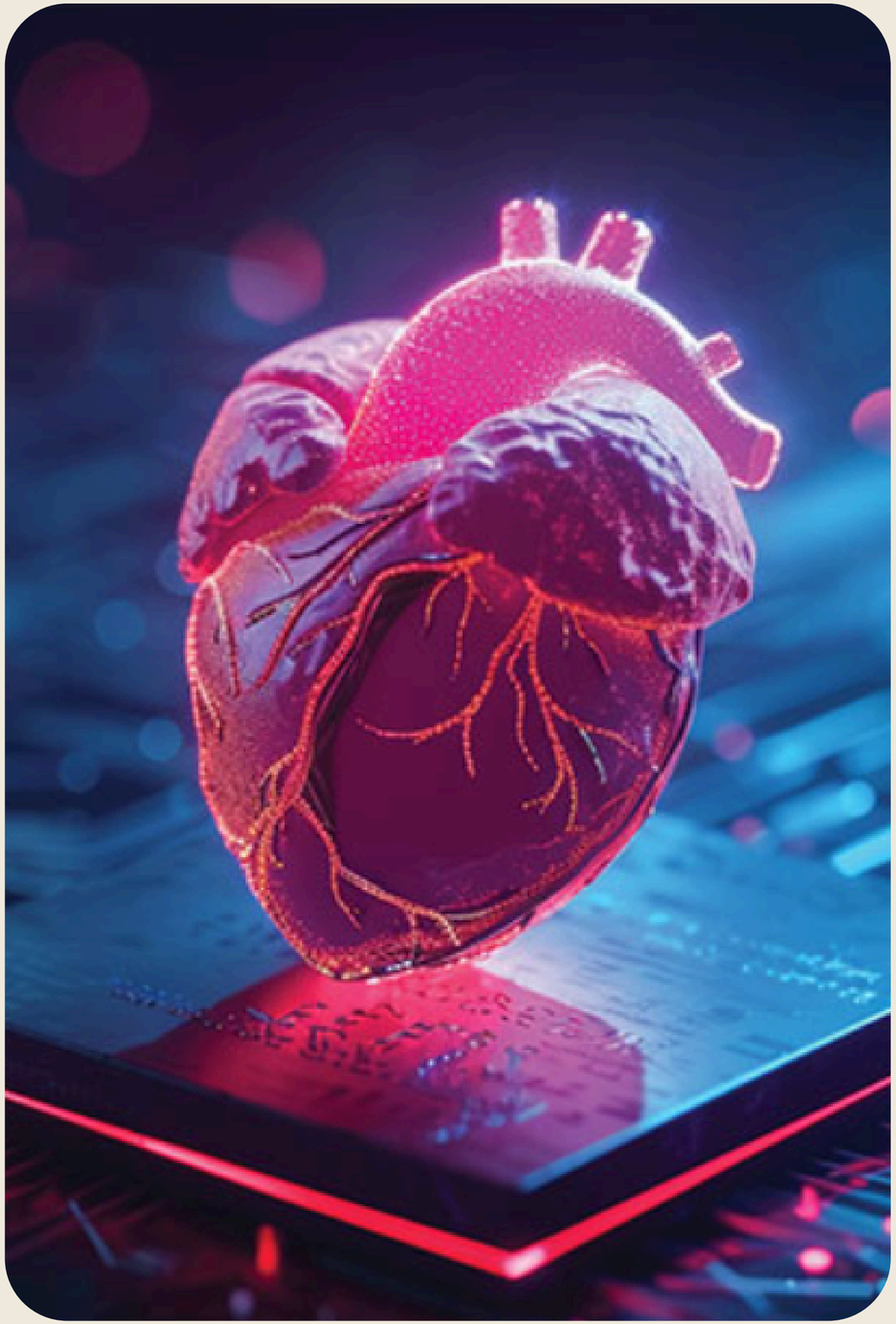
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External Advisor:

Rahman Sir



Problem Statement

The Problem

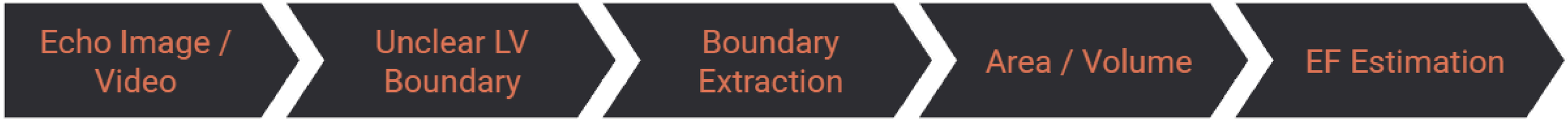
- EF estimation requires a reliable LV cavity boundary
- Accurate boundary extraction is essential for LV measurements

Challenges

- Speckle noise
- Low contrast
- Shadows and signal dropout
- Weak endocardial boundaries

Core Problem to Solve

- Reliable LV boundary extraction
- Convert boundary into area/volume
- Support accurate EF estimation
- Aim for a hardware-efficient solution



Target: LV cavity / endocardial border (blood–tissue interface).
Problem: develop a lightweight, interpretable and hardware-oriented pipeline for LV boundary extraction and EF estimation.
Scope: prototype development.

Existing Approaches & Research Gap

Positioning the project against classical and deep-learning methods

Existing Approach

- **Manual Tracing: Doctors manually trace the LV boundary to estimate EF.**
- **Image Processing: Filtering and segmentation techniques extract the LV.**
- **Model-Based: Active contours and shape models detect LV boundaries.**
- **AI-Based: Deep learning enables automatic LV segmentation and EF estimation.**
- **Key Challenges: Noise, weak boundaries, artifacts, and patient variability.**
- **Industry Trend: Toward automated, real-time, AI-assisted echocardiography.**

Research Gap

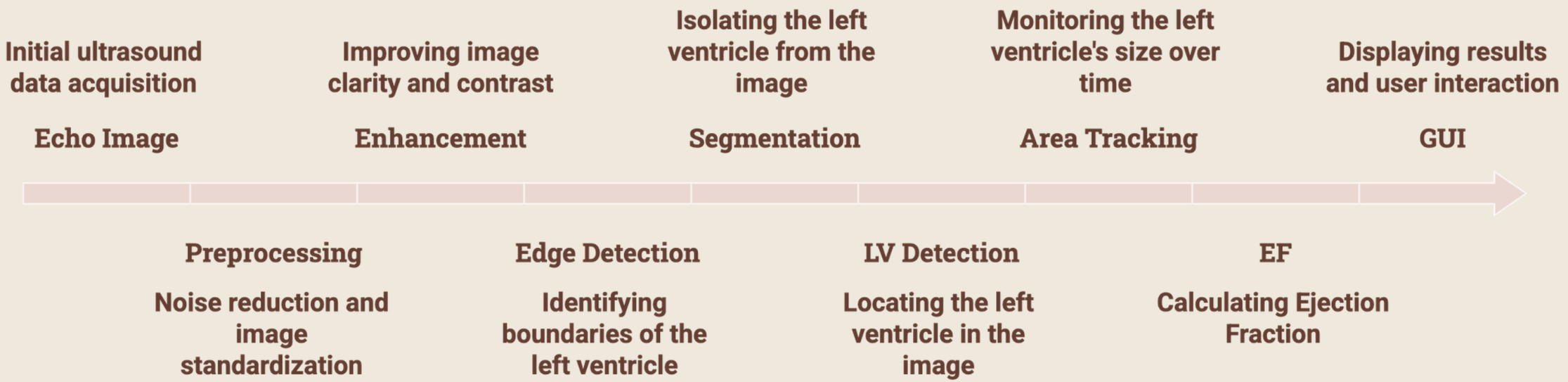
- Gap: Limited FPGA-based real-time LV segmentation compared with AI-based solutions.
- Our Edge: Hardware-accelerated classical image processing for low latency.
- No pretrained models: Interpretable and dataset-independent.
- End-to-end: LV segmentation → area/volume → EF estimation.
- Unique: Combines FPGA acceleration + medical image processing + standalone visualization in one system.

Proposed Solution: End-to-End Flow

From echocardiographic frame to LV contour and cardiac metric

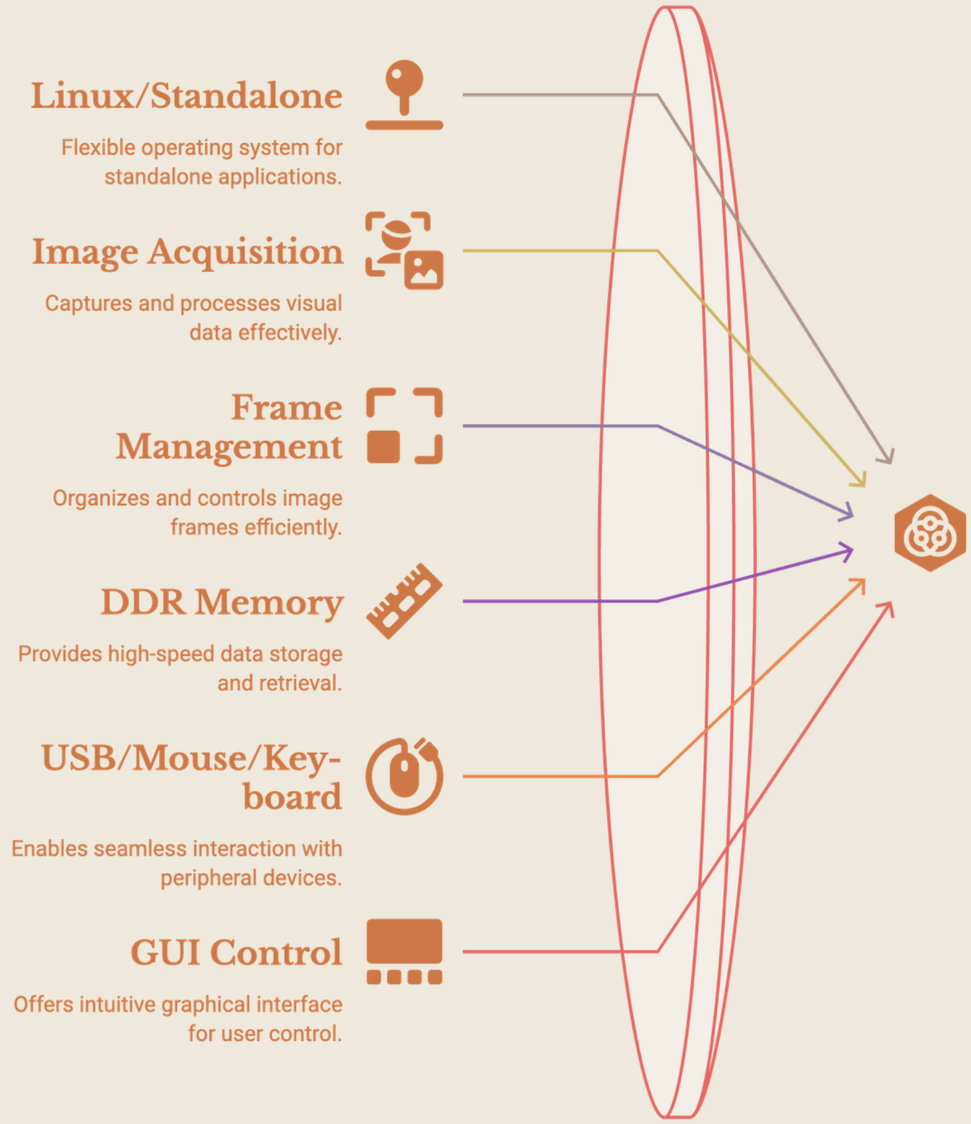
A lightweight, geometry-guided and FPGA-oriented pipeline for automated LV boundary extraction and ejection-fraction estimation from echocardiographic images/video.

Medical Image Processing Pipeline for Ejection Fraction



- Training-free / interpretable segmentation
- Anatomical geometry guides boundary extraction
- Designed for parallel FPGA implementation

FPGA System Architecture



Linux/Standalone

Flexible operating system for standalone applications.

Image Acquisition

Captures and processes visual data effectively.

Frame Management

Organizes and controls image frames efficiently.

DDR Memory

Provides high-speed data storage and retrieval.

USB/Mouse/Key-board

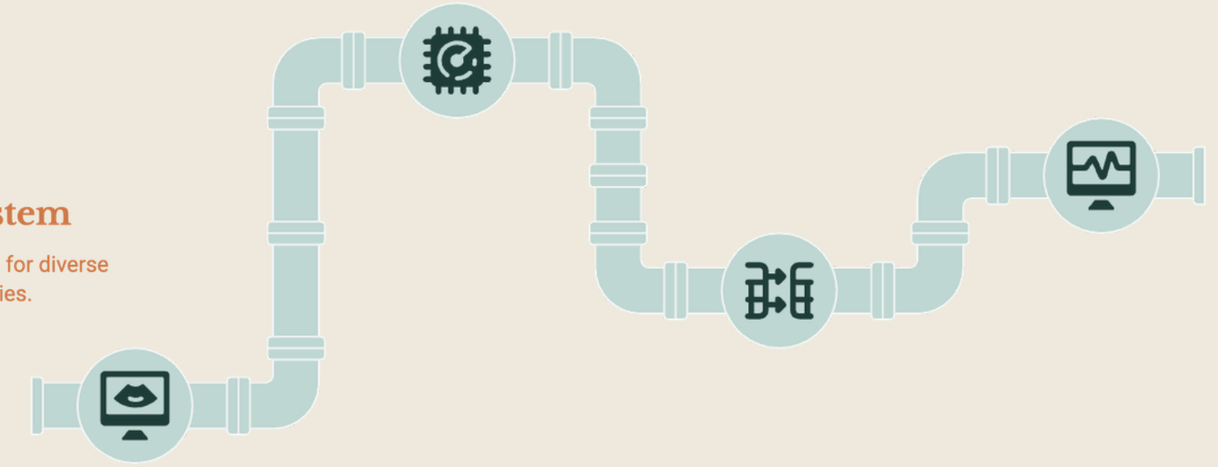
Enables seamless interaction with peripheral devices.

GUI Control

Offers intuitive graphical interface for user control.

Unified System

Integrated platform for diverse system functionalities.



01

Image and Video Input

Software on ARM Cortex-A9 handles image and video input, system control, GUI display, and frame management.

02

Hardware Acceleration

FPGA hardware performs preprocessing, filtering, enhancement, segmentation, LV detection, and area calculation.

03

Data Interconnect

AXI and DMA protocols facilitate communication between system components.

04

Output Generation

The system produces area tracking, EF estimation, and GUI updates.

Made with Ni

Methodology

End-to-end workflow for LV segmentation and quantitative cardiac analysis

01 Image Preprocessing

Morphological Reconstruction

- Smooths echocardiographic image
- Improves gray-level distribution
- Preserves useful structural information

FPGA-friendly local-window operation

02 FRFCM Segmentation

Histogram-Based Clustering + Membership Filtering

Clusters gray levels using FRFCM
Uses histogram representation to reduce computation
Filters membership to correct misclassified pixels

Training-free, interpretable segmentation

03 Geometry-Guided Boundary

Landmarks → LV Model → Refinement

P1 = Apex
P2 = Base Left | P3 = Base Right
$M = (P2 + P3) / 2$
$a = \|M - P1\|$ | $b = \|P3 - P2\| / 2$

Initial half-ellipse acts as an anatomical prior

04 Quantitative Analysis

LV Boundary → EDV / ESV → EF

Obtain refined LV boundary
Estimate LV area / volume at end-diastole and end-systole
Compute ejection fraction

EF = (EDV − ESV) / EDV × 100

Work Completed So Far

Research Papers

- Studied research papers related to echocardiography, LV segmentation, FCM and FRFCM
- Analysed existing approaches and identified their limitations

MATLAB Simulation

- Implemented and tested the image processing pipeline in MATLAB
- Worked on FRFCM-based segmentation
- Performed image cleaning and LV boundary extraction
- Developed the geometry-assisted boundary modeling approach

Demo Prototype

- Designed the PS–PL architecture and defined the data flow for the proposed FPGA-based image-processing pipeline
- Demonstrated that the proposed processing pipeline can be mapped to an FPGA-based architecture

Github Documentation

- **https://github.com/Sarveshkk05/EchoFCM-FPGA**
- Created and organised the project repository
- Documented the project workflow, algorithms and implementation progress

LV Segmentation Pipeline (MATLAB)

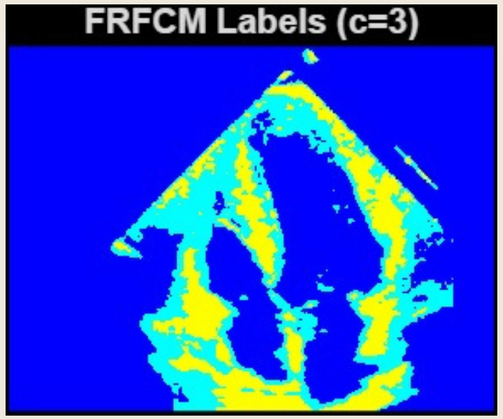
FRFCM Clustering

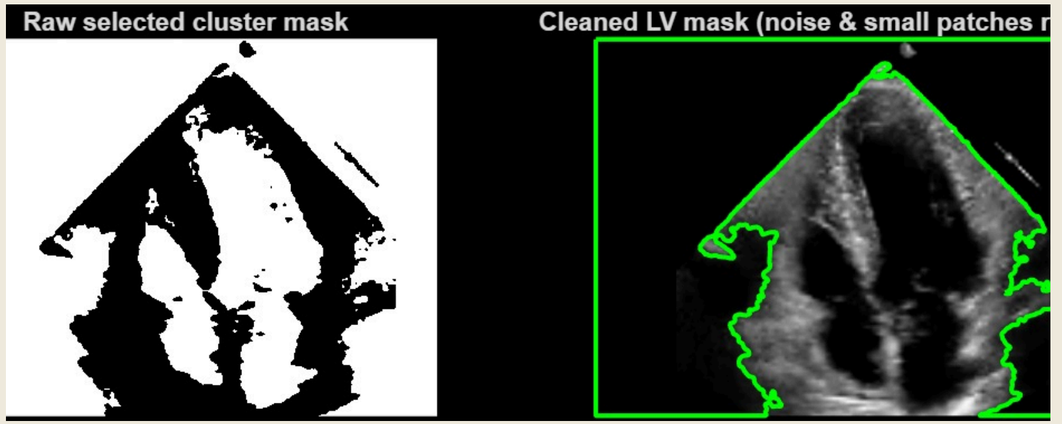
MorphologicaL Cleaning

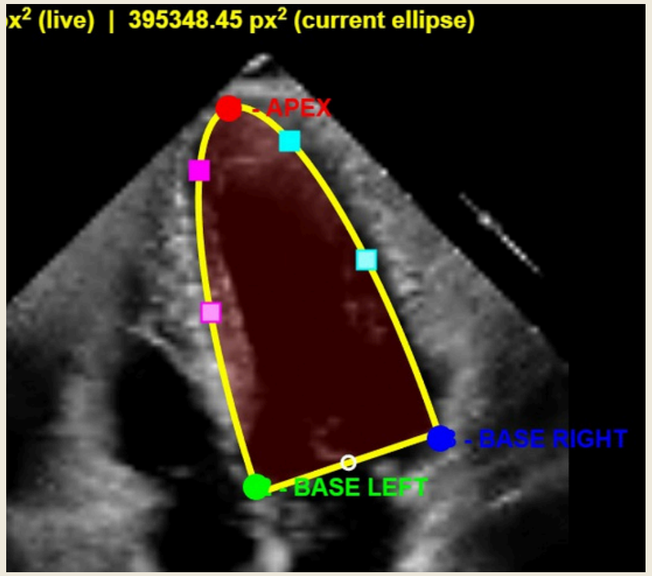
Interactive 3-Point Prolate Ellipse

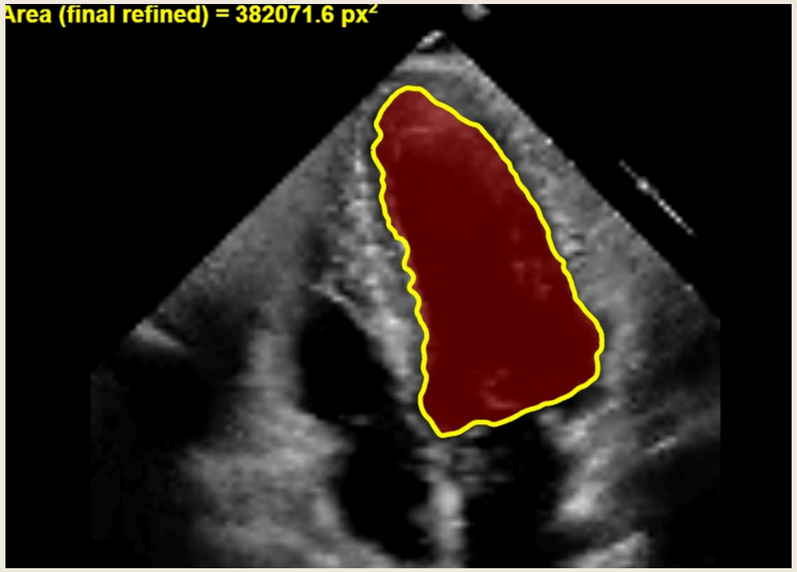
Active Contour Refinement

Final Overlay And Area calculation





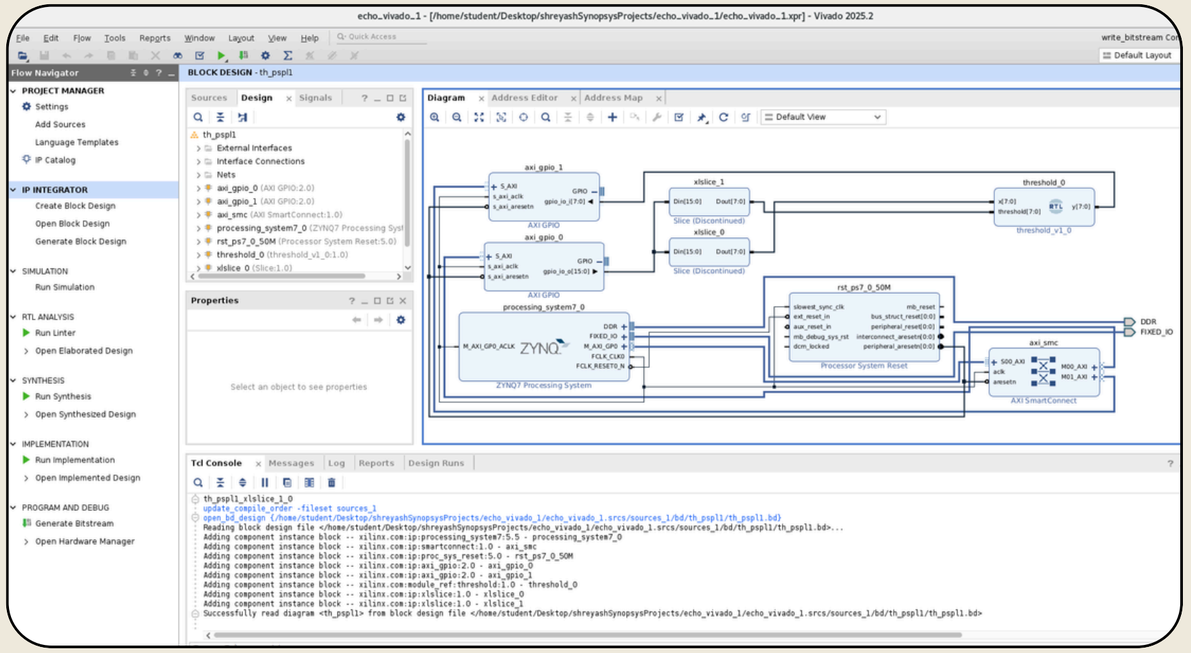


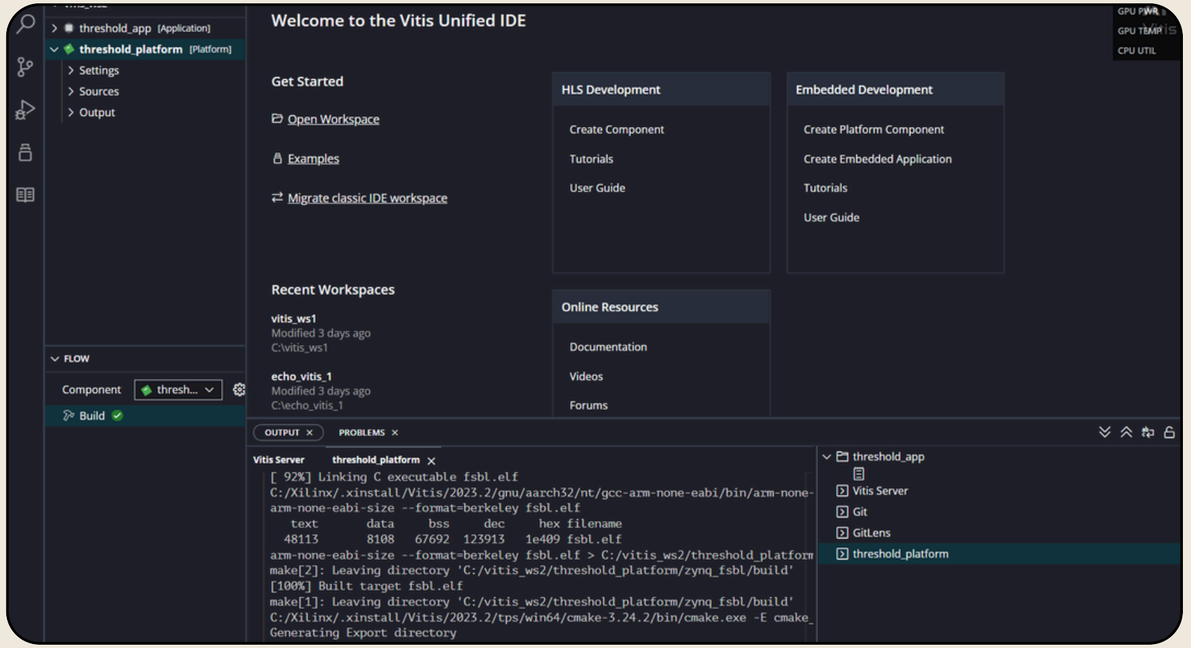


Hardware Design

Hardware Architecture

MATLAB: Algorithm & Verification
Vivado: Hardware Architecture
PS: ARM-based Control
PL: Hardware Processing
Synthesis: RTL to Hardware
Implementation: Placement & Routing
Vivado
↓
Bitstream (.bit)
↓
Vitis
↓
Hardware Platform
↓
Hardware Application
↓
FPGA Board
↓
System Execution





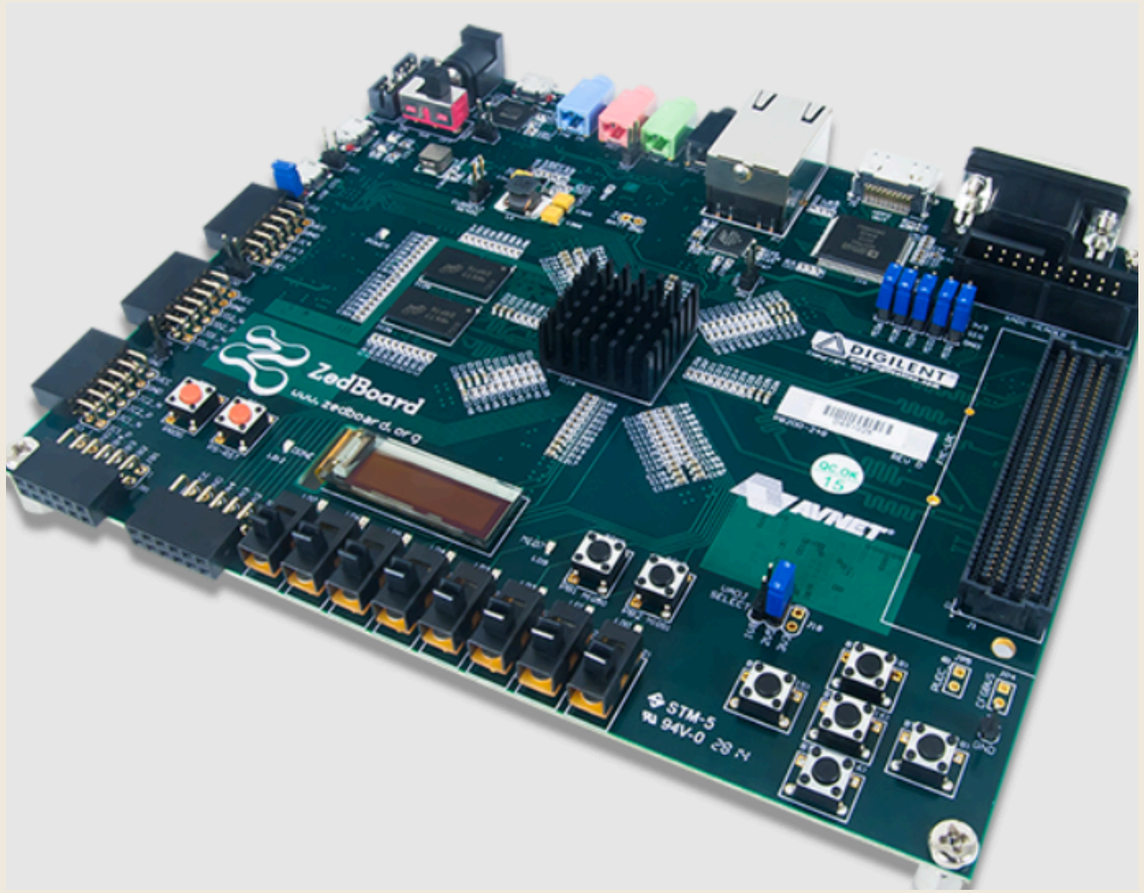
FPGA Hardware Deployment & Execution

VIVADO
Hardware Design
↓
Bitstream (.bit)
↓
VITIS
↓
Hardware Platform
↓
Hardware Application
↓
ZYNQ FPGA

PS
ARM CPU

PL
FPGA Logic

AXI
↓
System Execution



FUTURE WORK

1 SOFTWARE VALIDATION

- Complete FRFCM-based LV segmentation
- Validate with echocardiographic data & ground truth

2 CARDIAC METRICS

- Refine LV boundary extraction
- Extend to EDV, ESV & EF

3 FPGA IMPLEMENTATION

- RTL implementation
- Fixed-point optimization

4 HW–SW INTEGRATION

- FPGA + ARM integration
- Software–hardware verification

5 GUI IMPLEMENTATION

- Develop Qt/QML-based GUI
- Visualize LV contour & cardiac measurements

6 STANDALONE DEVICE

- Real-time processing
- Portable point-of-care system

References

• Moradi et al. (2019), MFP-Unet for left ventricle segmentation in echocardiography.

• Leclerc et al. (2019), Deep Learning for Segmentation Using an Open Large-Scale Dataset in 2D Echocardiography (CAMUS).

•Giorgio, A., Guaragnella, C., & Rizzi, M. (2023). FPGA-Based Decision Support System for ECG Analysis. Journal of Low Power Electronics and Applications, 13(1), 6.

•Lei et al. (2018), “Significantly Fast and Robust Fuzzy C-Means Clustering Algorithm Based on Morphological Reconstruction and Membership Filtering,” IEEE TFS.

• Lang et al. (2015), ASE/EACVI Recommendations for Cardiac Chamber Quantification by Echocardiography in Adults.

Thankyou